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MATHEMATICS
SHORT ANSWER PROBLEMS

Name : _____ **Index Number** : _____

Country : _____

17th International Mathematics and Science Olympiad

Indonesia

21 January 2021

Instructions:

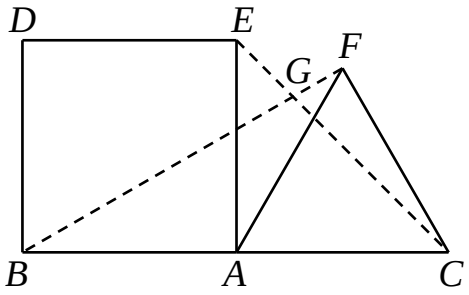
1. Write your name, country and index number on both the Question Booklet and the Answer Sheet.
2. Write your Arabic Numerical answers or English alphabet answers only in the Answer Sheet.
3. There are 25 questions in this paper.
4. For problems involving more than one answer, marks are only awarded when ALL answers are correct.
5. Each question is worth 1 mark. There is no penalty for a wrong answer.
6. You have 60 minutes to complete this paper.
7. Use black pen or blue pen or pencil to write your answer.

Do not turn over this page until you are told to do so.

SHORT ANSWER PROBLEMS

1. In year 2021, Ada's little brother is 10 years old. If Ada's age in 2021 is the sum of all the digits of the year she was born in. How old is she in 2021?

2. Points *A*, *B*, and *C* are on a straight line such that $AB = AC = 1$. Square *ABDE* and equilateral triangle *ACF* are drawn on the same side of line *BC* and lines *EC* and *BF* intersect at *G*. What is the measure, in degrees, of angle *EGB*?



3. If the twelve-digit number $\overline{155a710b4c16}$ is divisible by 396, where *a*, *b* and *c* are three distinct digits. What is the value of $a+b+c$?

4. Each cell of the 3×4 grid below contains a positive integer less than 9 such that the sum of the four numbers in each row are all equal, while the sum of the three numbers in each column are all equal as well. What number must be placed in the gray cell?

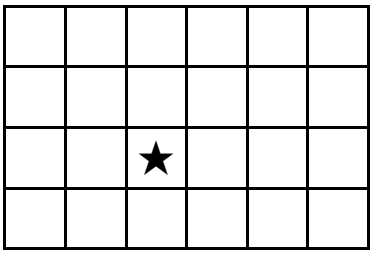
5	4		2
		3	6
3	5		

5. A computer store sells four types of laptop (A, B, C and D) and have sold a total of 2021 units. It is known that 20 type A laptops cost as much as 7 type B laptops, 4 type A laptops cost as much as 1 type C laptop and 4 type A laptops costs as much as 3 type D laptops. If the same amount was collected from each type, how many type A laptops were sold?

6. In the expression $(e\times7)\oplus f = 19$, it is known that "$\oplus$" can be any one of the following operations: $+$, $-$, $\times$ or $\div$ and *e* and *f* are both one-digit non-zero positive integers. What is the sum of all possible values of $(e+f)$?

7. The first 999 positive integers are listed on a board. In each turn, Peter selects any two numbers from the list and replace it with their positive difference. He keeps on doing this until only one number is left. What is the largest possible value of this number?

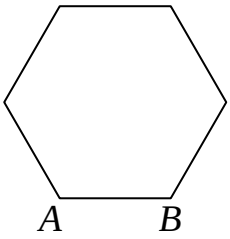
8. In the diagram below, how many different squares can be formed which contains the unit square with the "★" inside it?



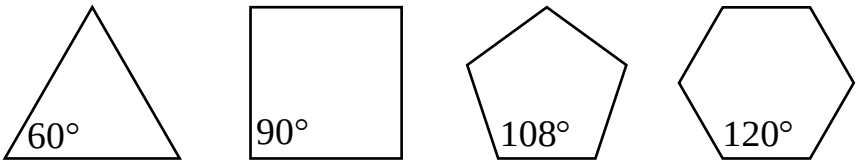
9. Cammy listed down all possible distinct five-digit positive even integers that can be formed using each of the digits 1, 3, 4, 5 and 9 exactly once. What is the sum of all the integers on Cammy's list?

10. Calculate the product of all the positive divisors of 120: $1\times 2\times 3\times 4\times ...\times 120$. It is known that this product is divisible by some 2^k where k is an integer. What is the largest possible value of k?

11. In the diagram below, Andy used a metal rod to form a regular hexagon where the length of side AB is equal to 16 cm. He then divides the hexagon into two equal parts where one part is used to construct a square, while the other part is used to construct an isosceles triangle with a base of 18 cm. What is the sum of the areas, in cm^2, of the newly created square and triangle?

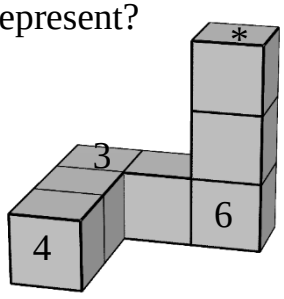


12. A regular triangle, a regular quadrilateral, a regular pentagon and a regular hexagon each have interior angles measured in integer degrees.



In total, how many kinds of regular polygons have interior angles measured in integer degrees?

13. The numbers 1 to 6 are printed on the six faces in each of the seven cubes and it is known that the sum of the two integers printed on any two opposite faces is 7. Arrange the seven cubes as shown in the diagram below such that the sum of the integers on any two faces touching each other is 8. What number does the "*" on the face of the top-most cube represent?

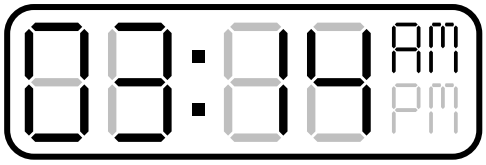


14. Seven points divide a circle into seven equal arcs. How many obtuse triangles can be formed by selecting three of these seven points as its vertices?

15. Positive integers from 1 to 45 inclusive are divided into 5 groups having 9 number each. What is the largest possible average of the medians of these 5 groups?
(Note: The median of a finite set of numbers is the "middle" number, when the numbers are listed from smallest to greatest or vice versa.)

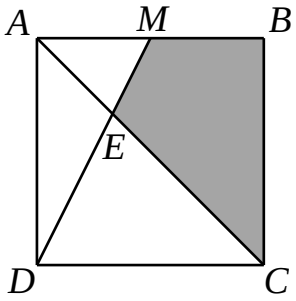
16. Find the smallest possible positive integer that is divisible by 18 which also has 18 different divisors?

17. I have a digital clock like the diagram shown below. During the 12-hour duration from 03:00 AM to 2:59 PM, how many minutes does the number "3" appear at least once?

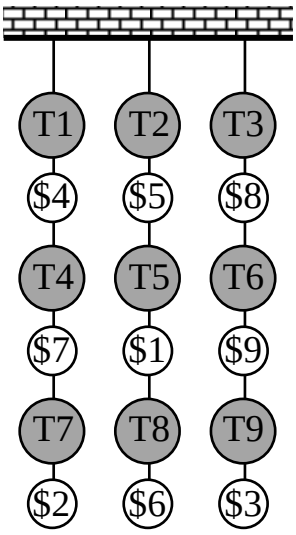


18. There are 32 coins such that all of them have a different weight. At least how many weighings on a standard 2-pan balance are needed to be able to determine both the heaviest and the second heaviest coins?

19. In the diagram below, *ABCD* is a square where *M* is the midpoint of *AB*. If line *AC* intersects *DM* at point *E* and the area of quadrilateral *BCEM* is 400 cm^2, then what is the area, in cm^2, of square *ABCD*?



20. In a target-shooting gallery, nine prizes ranging from \$1 to \$9, are placed and hanged up on three strings, as shown in the diagram below. The objective of the game is to hit the targets (indicated by the shaded circles) above each prize. If a target is hit, everything below it will be won by the player (and all the targets and prizes below becomes unavailable for the next player). It is known that the order of play was Alice went first, followed by Brian and then Colin. If each of them hits two targets and in total, Alice got \$18, Brian \$13 and Colin \$14, which target(s) were hit by Colin?



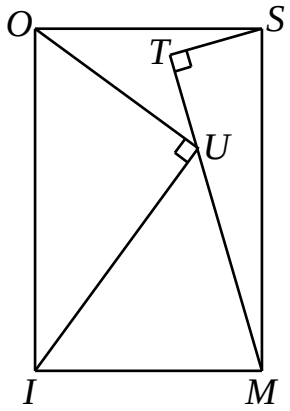
21. The following chart shows the answers given by Aaron, Betty, Clara and David in a ten-question true-or-false test. It is known that Aaron has eight, Clara seven and Betty only two correct answers. How many correct answers does David have?

Questions	1	2	3	4	5	6	7	8	9	10
Aaron	T	F	F	T	F	F	T	T	F	T
Betty	F	T	F	F	F	T	F	T	F	F
Clara	T	F	T	T	T	F	T	T	T	T
David	F	F	T	F	T	F	F	F	T	F

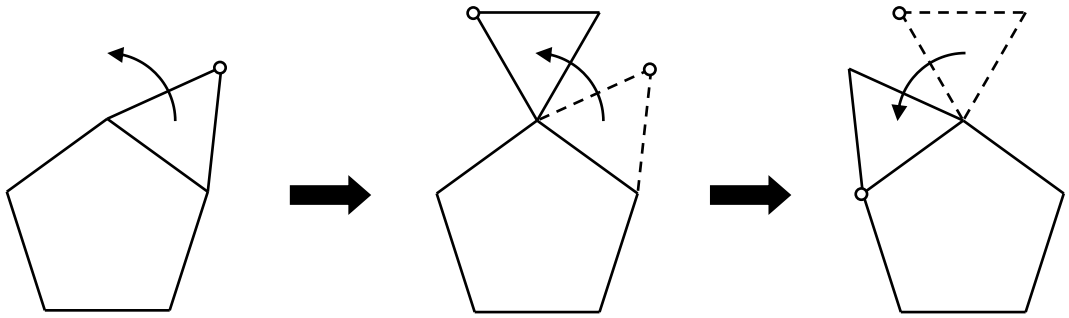
22. A group of 101 students went out for a school trip. Each of them visited at least one but at most two places out of the four places that were planned to be visited that day. Let *N* be the maximum number of students that visited the exact same places on the trip (e.g. they all visited only one place and this was the same place or they visited exactly two of the same places). What is the minimum value of *N*?

23. The average of $\overline{abc}$, $\overline{ab}$, $\overline{bc}$ and $\overline{ca}$ is *M*. If we remove $\overline{ab}$, then the average of the three remaining numbers will become *N*. If it is known that the difference between *N* and *M* is 40, then what is the maximum possible value for $\overline{abc}$?

24. In the diagram below, *STM* and *OUI* are right triangles inside rectangle *IMSO* such that point *U* lies on line *TM*. If $ST = 21\,\text{cm}$, $TM = 72\,\text{cm}$ and $OU = 45\,\text{cm}$, what is the area, in cm^2, of rectangle *IMSO*?



25. An equilateral triangle rotates around a regular pentagon of side 21 cm as shown in the diagram below.



Original position

What is the length, in cm, of the trace made by the white point until it first returns to its original position? (Let $\pi = \frac{22}{7}$)

17th International Mathematics and Science Olympiad

SHORT ANSWER PROBLEMS

ANSWER SHEET

Name : _____ **Index Number** : _____

Country : _____ **Score** : _____

1.	years old	***14.***	obtuse triangles
2.	°	***15.***	
3.		***16.***	
4.		***17.***	minutes
5.	laptops	***18.***	weighings
6.		***19.***	cm^2
7.		***20.***	and
8.	squares	***21.***	correct answers
9.		***22.***	
10.		***23.***	
11.	cm^2	***24.***	cm^2
12.	kinds	***25.***	cm
13.			